Contents

1 Defintions

Lets list some definitions of Tensor. A Tensor T can be regarded as an element of a tensor product space

$$T \in \bigotimes_{j\in J} V^j$$

and we say T is a $|J|$-legged tensor.

Given finite dimensional Vector spaces V and W, we have the identity that $V \otimes W^* \equiv Hom(W, V)$. Now, if we have a Tensor $T \in V_1 \otimes \cdots \otimes V_n$, and since for any Vector space V, $(V^*)^* = V$, we can peel off any number of V_is, take their duals and represent T as a map

$$T : V_n^* \otimes \cdots \otimes V_{k+1}^* \to V_1 \otimes \cdots \otimes V_k$$

Note that we present the domain here in a contravariant fashion due to the contravariance of the dual.

Lets look at an example, let $t : V_1 \otimes V_2 \otimes W_2^* \otimes W_1^*$. We can represent as an operator in dirac notation:

$$t = \sum_{i_1,i_2,j_1,j_2} T_{j1j2i1i2} |v_{i_1}\rangle |v_{i_2}\rangle \langle w_{j_1}| \langle w_{j_2}|$$

Graphically, we represent a general tensor

$$T : (W_n^* \otimes \cdots \otimes W_1) \to (V_1 \otimes V_2 \otimes \cdots \otimes V_n)$$

as follows, where we choose incoming arrows for duals in the domain and outgoings for normal spaces, and on the bottom we choose incoming for duals and outgoing for normals.

1.1 Symmetric Tensors

Let G be a compact group. Consider a tensor T living in the tensor product space of its legs:

$$T \in V_1 \otimes \cdots \otimes V_n \otimes W_m^* \otimes \cdots \otimes W_1^*$$

Let $\rho_i : G \to \mathrm{GL}(V_i)$ be the linear representation of G on each vector space V_i, and let $\sigma_j^* : G \to \mathrm{GL}(W_j^*)$ be the dual representation on W_j^*.

The total action of the group G on the entire tensor product space is given by

$$\Pi : G \to \mathrm{GL}\left(\bigotimes_i V_i \otimes \bigotimes_j W_j^*\right)$$

$$\Pi(g) = \rho_1(g) \otimes \cdots \otimes \rho_n(g) \otimes \sigma_m^*(g) \otimes \cdots \otimes \sigma_1^*(g)$$

A tensor T is defined to be a **symmetric tensor** (or G-invariant tensor) if it is invariant under simultaneous action on all legs for all $g \in G$ [SPV10]:

$$\Pi(g)T = T \quad \forall g \in G$$

By interpreting T as a map

$$T : W_1 \otimes \cdots \otimes W_m \to V_1 \otimes \cdots \otimes V_n$$

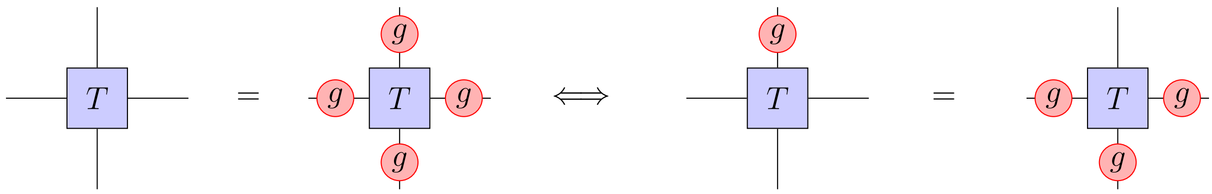
it follows directly from the definition of global invariance that the tensor must commute with the group action and we get the following definition of a symmetric tensor:

Definition 1.1 (symmetric Tensor)**.** A *symmetric tensor* is an intertwiner of the representations of the group G:

$$\bigotimes_{j=1}^{m} \sigma_j(g) T = T \bigotimes_{i=1}^{n} \rho_i(g) \quad \forall g \in G,$$

which we denote by

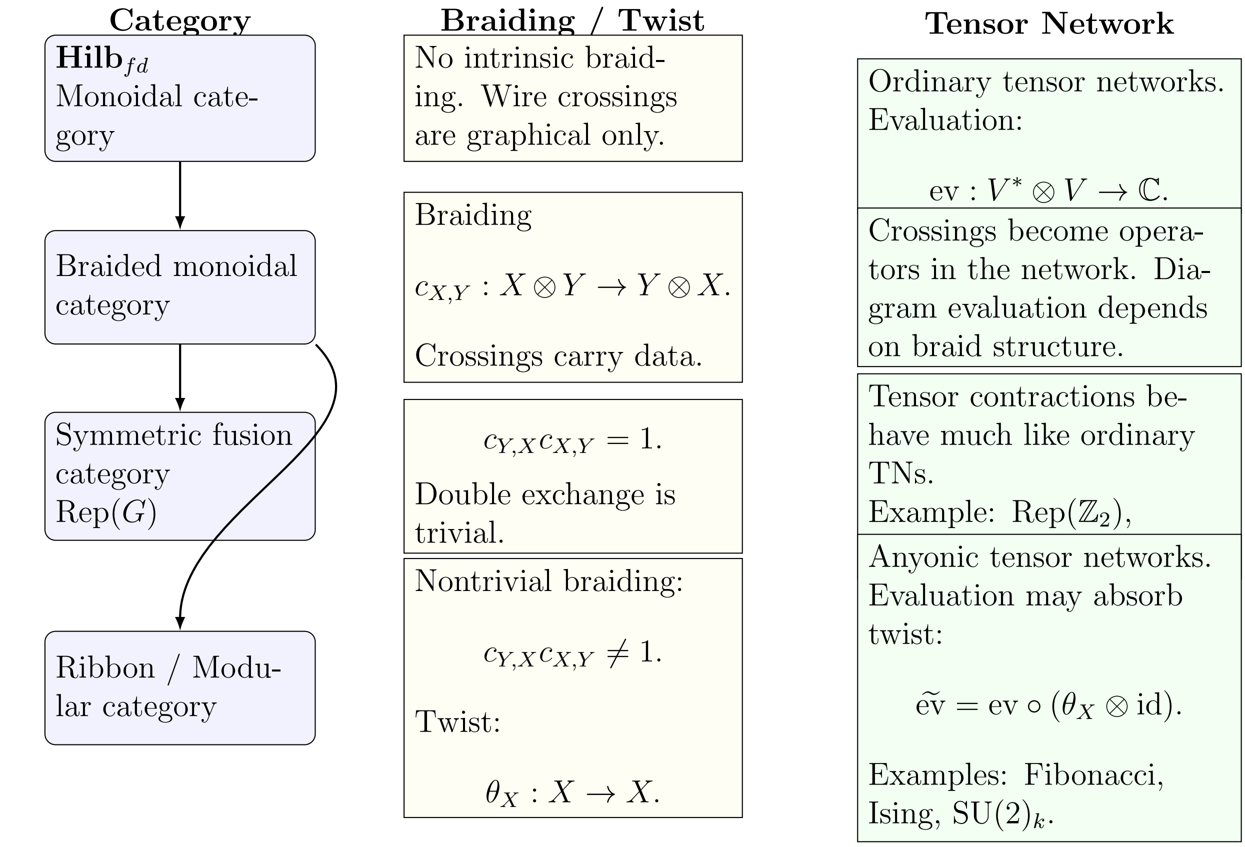
$$T \in Hom(W, V).$$



Corollary 1.1.1 (Networks of symmetric tensors are symmetric)**.** Since a tensor network is built by composition and application of the $(co)ev$ maps, both of which are intertwiners, the full network is also an intertwiner. Important there is that this entire discussion is only about FdVect or FdHilb for now, but i think the same applies to any braided category.

1.2 Review of Symmetric Tensor networks.

Generally, we follow the notations of [SV12], [Sch+20], [Sch+20], [Sch20] which gave an overview for implementing $SU(2)$ invariant tensor networks. A general overview for (Non)abelian symmetries is [SPV10], and a detailed account for the abelian $U(1)$ in [SPV11]. There are a multitude of implementations given in major packages, such as ITensor, [FWS22], Cyten, TenPy, [Hau+24], QSpace, [Wei24] ChemTensor, UniTensor, YasTn (Abelian Only) [Ram+25] (MPSKit?), (Cytnx (Abelian only i think.))). A general account of symmetries is given in [Unf24] with its respective implementation in [Hau+24] (We have to build cytens citation manually.)



2 Symmetries

What do we know now? First of all, we take a symmetry group G that is finite and compact (why?)[For compact groups, representations are completely reducible (Maschke) and the irreps are finite aswell. Finite groups are contained in compact groups.]. Then we look for its unitary representation $U(G)$ on a Hilbert space H. Furhtermore, if this is a multiparty hilbert space, lets say n sites, we view the unitary representations U_n at site n. The unitary representation $U_n : G \to Hom(V_n, V_n)$ is a covariant linear unitary map. A map $V_i \to V_j$ is symmetric if it is an intertwiner of the representations, i.e. the following diagram commutes:

$$\begin{tikzcd} V_i \arrow[r, "U_{V_i}(g)"] \arrow[d, "f"'] & V_i \arrow[d, "f"] \\ V_j \arrow[r, "U_{V_j}(g)"'] & V_j \end{tikzcd}$$

Note that here V_i is called a *symmetry space*.

Some examples of these groups include $\mathrm{U}(1), \mathrm{SU}(2), \mathbb{Z}_n$

Now where the intutition is more shaky: Consider a symmetry space V_i on a single site (why single). Then we get that it decomposes into a direct sum of irreps V_i^a,

$$V_i = \bigoplus_a \bigoplus_{i=1}^{d_a} V_i^a$$

and we can easily construct an isomporphism (but this depends on choosing a basis for D_a.)

$$V_i = \bigoplus_a \bigoplus_{i=1}^{d_a} V_i^a = \bigoplus_a (D_a \otimes V_i^a).$$

Well, the isom we can construct here would look like $\bigoplus_1^{d_a} V_i^a \to D_a \otimes V_i^a$.

Consider a single site of 3 spin 1/2 particles. Under SU(2) this decomposes as

$$2 \otimes 2 \otimes 2 = 4 \oplus 2 \oplus 2$$

Now consider that C^2 is just the fundamental representation of SU(2).

Note here, an invariant subspace $W \subseteq V$ is a subset of vectors s.t. for all $U \in \mathrm{SU}(2)$, $Uw \in W$.

symmetry obviously implies pullthrough properties, do Intertwiners as well?

Yes, because $U_(V \otimes W) : g \to U_V(g) \otimes U_W(g)$ and the following diagram commutes:

$$\begin{tikzcd} V_i \otimes V_j \arrow[r, "U_{V_i \otimes V_j}(g)"] \arrow[d, "T"'] & V_i \otimes V_j \arrow[d, "T"] \\ V_k \arrow[r, "U_{V_k}(g)"'] & V_k \end{tikzcd}$$

2.1 Some Symmetry Groups

2.1.1 $\mathbb{Z}_n$

This group is given by integers $0, \ldots n-1$ with addition mod n. Since this happens to be abelian, all the irreps are 1 dimensional, i.e. $U_a(g) : g \to Hom(\mathbb{C}, \mathbb{C})$ with representatives $U_a(g) = z \mapsto e^{2\pi i a g/N} z$. What isthe fusion rule here?

$$a \otimes b \equiv (c := a + b \mod n)$$

This also does not have fusion multiplicities.

2.1.2 The group SU(2)

Summarize the results from the paper here.

2.1.3 The group $\mathrm{SU}(2)_3$

Technically this is a fusion category, the thesis [Unf24] handles this distinction. But if we think of this in terms of the programming guide, this seems to be doable using a group.

Now, since we can argue that Products of symmetry groups again form a symmetry group (proof?), we would also need another type for sector.

We could use $classSector(NamedTuple)$ holding an array charges.

2.2 Notes on structural tensors and fusion trees

What do the CG tensors look like? As they are three index and have no intermediates, these are simply $Q_{j_1,j_2,j_3} = (C^{fuse})^{(j_3,m_{j_3})}_{(j_1,m_{j_1}),(j_2,m_{j_2})}$.

2.3 Notes on the papers

Lets assume we only look at incoming legs, i.e. we look at T_{j_a,j_b,j_c} in a composite vectors space $V^a \otimes V^b \otimes V^c$. By the decomposition

$$V^a \otimes V^b \otimes V^c = \left(\bigoplus_{j_a} D_{j_a} \otimes V_{j_a}\right) \otimes \left(\bigoplus_{j_b} D_{j_b} \otimes V_{j_b}\right) \otimes \left(\bigoplus_{j_c} D_{j_c} \otimes V_{j_c}\right)$$

and we can represent a single index inside the block space

$$T_{j_a,j_b,j_c} \in (D_{j_a j_b j_c} \otimes V_{j_a j_b j_c})$$

Choosning a basis $t = (t_{j_a}, t_{j_b}, t_{j_c})$ and $m = (m_{j_a}, m_{j_b}, m_{j_c})$ we can write

$$T_{j_a,j_b,j_c} = \sum_{t,n} (T_{j_a,j_b,j_c})_{t,m} (|t\rangle \otimes |m\rangle)$$

Now we can use Wigner-Eckart to obtain

$$T_{j_a,j_b,j_c} = \sum_{t,n}((P_{j_aj_bj_c})_t \cdot (Q_{j_aj_bj_c})_m)_{t,m}(|t\rangle \otimes |m\rangle)$$

$$= \left(\sum_t (P_{j_aj_bj_c})_t \, |t\rangle\right) \otimes \left(\sum_m (Q_{j_aj_bj_c})_m \, |m\rangle\right) = P_{j_aj_bj_c} \otimes Q_{j_aj_bj_c}$$

If we generalize this with the decomposition for k indices $i = (i_1, i_2, \ldots, i_k)$ and let $j_e = (j_{e_1}, j_{e_2}, \ldots, j_{e_{k-3}})$ be the internals

$$T_i = \sum_{j_e}(P_j^{j_e})_t \cdot (Q_j^{j_e})_m$$

2.4 Notes on the decomposition

Suppose that (V, ρ_V) and (W, ρ_W) are representations of G.

Definition 2.1. A linear map $f : V \to W$ is called an *intertwiner* if

$$f \circ \rho_V(g) = \rho_W(g) \circ f \quad g \in G.$$

Define the *Vector space of Interwtiners* as

$$Hom_G(V, W)$$

Suppose now V^a is one of the reps that decomposes as

$$V^a \eqsim \bigoplus_{\cdot=1}^{d_a} R_a.$$

Lemma 2.2. For a vector space W, there is an isomorphism

$$\bigoplus^m W \simeq \mathbb{C}^m \otimes W.$$

Proof. Let $e_{i i\in[m]}$ be the standard basis on $\mathbb{C}^m$. Let

$$\Phi : \mathbb{C}^m \otimes W \to \oplus^m W$$

be defined by

$$\phi(e_i \otimes w) = (0, \ldots, w, \ldots 0)$$

and linear, thus

$$\Phi\left(\sum_i c_i e_i \otimes w_i\right) = (c_1 w_1, \ldots, c_m w_m).$$

is clearly bijective, and we have our desired isomorphism. ∎

Lemma 2.3. Let D be a finite dimensional k-vector space and let V be an object of a k-linear category C. Then

$$D \otimes V \cong V^{\oplus \dim D}$$

Proof. Let $n = \dim D$. Take a basis $\{e_i\}$ for D. This yields an isomorphism

$$D \cong \bigoplus_{i=1}^{n} k \cdot e_i \cong k^n$$

Since in a k-linear category the tensor product is additive (should we prove this?), we can distribute

$$D \otimes W \cong \bigoplus_{1}^{n} (k \cdot e_i \otimes W) \cong \bigoplus W = W^{\oplus n}$$

□

Lemma 2.4 (Isotopic Decomposition)**.** Let C be a semisimple k-linear category with simple objects $\{R_a\}$. Then every object $V \in C$ admits a canonical decomposition

$$V \cong \bigoplus_{a} D_a \otimes R_a. \tag{1}$$

where we refer to $D_a \coloneqq \mathrm{Hom}_C(R_a, V)$ as the *degeneracy* of the space.

Proof. Since C is semisimple, there exist integers $m_a \geq 0$ such that

$$V \cong \bigoplus_{a \in L} R_a^{\oplus m_a}.$$

Applying the additive functor $\mathrm{Hom}_C(R_b, -)$ yields

$$\mathrm{Hom}_C(R_b, V) \cong \bigoplus_{a \in L} \mathrm{Hom}_C(R_b, R_a^{\oplus m_a}) \cong \bigoplus_{a \in L} \mathrm{Hom}_C(R_b, R_a)^{\oplus m_a}.$$

Then, by Schurs lemma

$$\bigoplus_{a \in L} Hom_C(R_b, R_a)^{\oplus m_a} = k^{\oplus m_b}$$

□

So we actualy need this: Maybe this is simpler if we always let the degeneracy be C?

Lemma 2.5. Let C be a k-linear category and let D, E be finite-dimensional k-vector spaces. For any objects $R, S \in C$, there is a natural isomorphism

$$\mathrm{Hom}_C(D \otimes R, E \otimes S) \cong \mathrm{Hom}_k(D, E) \otimes \mathrm{Hom}_C(R, S).$$

Proof. Let $m = \dim_k D$ and $n = \dim_k E$. By unfolding the definitions and utilizing the properties of finite biproducts, we establish the result through a direct chain of natural isomorphisms:

$$\begin{aligned}
\mathrm{Hom}_C(D \otimes R, E \otimes S) &\cong \mathrm{Hom}_C(R^{\oplus m}, S^{\oplus n}) && \text{by 2.2} \\
&\cong \bigoplus_{i=1}^{m} \bigoplus_{j=1}^{n} \mathrm{Hom}_C(R, S) && \text{by additivity of } \mathrm{Hom}_C \\
&\cong k^{mn} \otimes \mathrm{Hom}_C(R, S) && \text{by 2.2} \\
&\cong \mathrm{Hom}_k(D, E) \otimes \mathrm{Hom}_C(R, S) && \text{since } \mathrm{Hom}_k(D, E) \cong k^{mn}
\end{aligned}$$

(The last as s) □

Theorem 2.6 (Decomposition of Semisimple Categories)**.** Let C be a semisimple, monoidal, k-linear category whose tensor product is k-linear in each variable, with simple objects $\{R_a\}_{a \in L}$ and let V_i, W_i be a collection of objects in C. Every morphism

$$T \in \mathrm{Hom}_C\left(\bigotimes_i V_i, \bigotimes_i W_i\right)$$

admit a decomposition as

$$T = \bigoplus_{\hat{a},\hat{b}} P^{(\hat{a},\hat{b})} \otimes Hom_C(R_{\hat{a}}, R_{\hat{b}}),$$

where $\hat{a} = (a_1, \ldots, a_n)$ and P is a k-vector space of dimension ???

Proof.

$$\begin{aligned}
&\mathrm{Hom}_C\left(\bigotimes_i V_i, \bigotimes_j W_j\right) \\
&\quad \cong \mathrm{Hom}_C\left(\bigotimes_i \left(\bigoplus_{a_i} D_{i,a_i} \otimes R_{a_i}\right), \bigotimes_j \left(\bigoplus_{b_j} E_{j,b_j} \otimes R_{b_j}\right)\right) && \text{by 2.4} \\
&\quad \cong \mathrm{Hom}_C\left(\bigoplus_{\hat{a}} D_{\hat{a}} \otimes R_{\hat{a}}, \bigoplus_{\hat{b}} E_{\hat{b}} \otimes R_{\hat{b}}\right) && \text{since } \otimes \text{ is } k\text{-linear in each variable} \\
&\quad \cong \bigoplus_{\hat{a},\hat{b}} \mathrm{Hom}_C(D_{\hat{a}} \otimes R_{\hat{a}}, E_{\hat{b}} \otimes R_{\hat{b}}) && \text{by additivity of } \mathrm{Hom}_C \\
&\quad \cong \bigoplus_{\hat{a},\hat{b}} \mathrm{Hom}(D_{\hat{a}}, E_{\hat{b}}) \otimes \mathrm{Hom}_C(R_{\hat{a}}, R_{\hat{b}}) && \text{by 2.5.}
\end{aligned}$$

□

Example 1. If we restrict to the category **Rep**$(SU(2))$ we can build the symmetric tensor networks as given in [SV12]. TODO: actual ly do this and convince yourself that the notations amd the triplet basis workd as desired.

2.5 Q intertwiners

Lets look at 2 paths for the 4 index tensor, $(a, b, c \to d)$ and $(a, b \to c, d)$.

$$(Q^{j_e}_{j_a j_b j_c})_{m_1,m_2,m_3,m_4} = \sum_{m_e} \langle j_1 m_1; j_2 m_2 | j_e m_e \rangle \langle j_e m_e; j_3 m_3 | j_4 m_4 \rangle$$

while the other topology would look like

$$Q \ldots = \sum_{m_e} \langle j_1 m_1; j_2 m_2 | j_e m_e \rangle \langle j_e m_e | j_3 m_3; j_4 m_4 \rangle$$

and as an operator this would have to live in $Hom(V_{j_1} \otimes V_{j_2}, V_{j_3} \otimes V_{j_4})$

The yoga tree $(-1, 1, -3), (-2, 1, -4)$ Gives us a good mixture of those. This Q would then have to look like

$$Q^1_{-1-2-3-4} = \sum_1 \langle j_{-1} m_{-1}; j_1 m_1 | j_{-3} m_{-3} \rangle \langle j_{-2} m_{-2} | j_1 m_1; j_{-4} m_{-4} \rangle = \sum_{m_1} C_{-1,1\to -3} \cdot C_{-2,1\to -4}$$

How would we do this when we have a contraction of two simultaneous indices:

$$Q^{1,2}_{-1-2} = \sum_{1,2} \langle -1 | 1; 2 \rangle \langle 1; 2 | -2 \rangle$$

If we define τ to be the fusion trees topology, then we can define ${}_\tau Q^i_j$

What would this look like for general symmetry groups G? Lets choose a basis of its invariant decomp $|\lambda_a t_a \mu_a\rangle$. Then the generalized CG coefficient would look like

$$(C^\alpha_{\lambda_a \lambda_b \to \lambda_c})^{\mu_c}_{\mu_a \mu_b} = \langle \lambda_a \mu_a; \lambda_b \mu_b | \lambda_c \mu_c \rangle_\alpha$$

where alpha denotes the singular fusion multiplicity at μ_c.

Now to fully specify Q here for k sectors (λ_i, μ_i), we need to give it $k-3$ internal edges $(\lambda_{e_1}, \ldots, \lambda_{e_{k-3}})$ along with $k-2$ mulitplicity symbols, $(\alpha_1, \ldots, \alpha_{k-2})$

Can we write down a general formula for Q?

This would then be

$$Q_\lambda$$

3 Categories

3.1 Tensor and Fusion Categories

Let C be a category with bifunctor

$$\otimes : C \times C \to C, \quad (X, Y) \to X \otimes Y.$$

and the isomporphism $F : (- \otimes -) \otimes - \simeq - \otimes (- \otimes -)$

... introduce fusion categories here, with associativity and commutativity restrictions, I think this would be a good start.

Definition 3.1 (Monoidal Category)**.** A monoidal category is a category C with a bifunctor $\otimes : C \times C \to C$

Let C and D be linear abelian categories. Then the *Deligne Tensor Product* $C \boxtimes D$ is a linear abelian category with a bifunctor

$$\boxtimes : C \times D \to C \boxtimes D$$

with morphisms given by

$$Hom_{C \boxtimes D}(X \boxtimes Y, W \boxtimes V) = Hom_C(X, W) \otimes Hom_D(Y, V)$$

Definition 3.2 (The Deligne Tensor Product of Categories)**.** Let k be a field, and let $\mathcal{C}$ and $\mathcal{D}$ be two k-linear, finite abelian categories. The **Deligne tensor product**, denoted by $\mathcal{C} \boxtimes \mathcal{D}$, is a k-linear finite abelian category equipped with a k-bilinear functor

$$\boxtimes : \mathcal{C} \times \mathcal{D} \longrightarrow \mathcal{C} \boxtimes \mathcal{D}, \quad (X, Y) \mapsto X \boxtimes Y$$

TODOTOD!!

Definition 3.3 (The Category $\mathbf{Rep}_k(G)$)**.** Let G be a group and k a field. The *category of linear representations of G over k* consists of the following data:

- **Objects:** The objects $\mathbf{Rep}_k(G)$ are pairs (V, ρ_V), where V is a k-vector space and $\rho_V : G \to \mathrm{GL}(V)$.
- **Morphisms:** For any two objects (V, ρ_V) and (W, ρ_W) in $\mathbf{Rep}_k(G)$, a morphism from V to W is a k-linear transformation $f : V \to W$ that is G-invariant (also called an *intertwiner*). That is, for every element $g \in G$, the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{\rho_V(g)} & V \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f} \\ W & \xrightarrow[\rho_W(g)]{} & W \end{array}$$

In algebraic terms, f must satisfy the condition:

$$f \circ \rho_V(g) = \rho_W(g) \circ f \quad \forall g \in G$$

The set of all such morphisms is denoted by $\mathrm{Hom}_G(V, W)$.

We can further endow $Rep_k(G)$ with a monoidal structure: For any two objects $(V, \rho_V), (W, \rho_W)$, their tensor product is in $\mathbf{Rep}_k(G)$ by the diagonal action

$$\rho_{(V \otimes W)}(g)(v \otimes w) = \rho_V(g)v \otimes \rho_W(g)w$$

Theorem 3.4. Let G_1 and G_2 be finite groups and let $\mathbf{Rep}(G_1), \mathbf{Rep}(G_2)$ be their respective symmetry categories. Then there is an equivalence of categories

$$\mathbf{Rep}(G_1 \times G_2) \simeq \mathbf{Rep}(G_1) \boxtimes \mathbf{Rep}(G_2)$$

Big issue here: $\mathbf{Rep}(\mathrm{SU}(2))$ is not a fusion category as it is not finite. But if i read it correctly, the deligne product only requires abelianness of its groups. Now, $\mathbf{Rep}(G)$ is monoidal for any G but is it abelian?

So this is actually true, it is abelian. We could obtain a restricted versions by for example always restricting $SU(2)$ to some $SU(2)_k$ for some finite k. But i do not yet know how to present this nicely.

A second advantage would be that we can always represent fusion trees as monoidal categories (is this true or is fusion required here again?), and we can write down a unified theory for the fusion trees. Then I hope to write a nice method to view the "eval" of such a fusion tree, i.e. its clebsch gordan tensors by contracting down the topology of the tree.

4 Categories Needed

The motivation for this section is to introduce the necessary categoy theory background, and provide all the necessary theorems for a format that includes both representations and fusion categories.

Definition 4.1 (Monoidal Category)**.** A monoidal category is a category C equipped with

1. A bifunctor $\otimes : C \times C \to C$.
2. A unit element $1 \in C$
3. An isomorphism
$$\alpha : - \otimes (- \otimes -) \to (- \otimes -) \otimes -$$
4. Left and right unitors $\gamma : 1 \otimes - \to -$ and $\delta : - \otimes 1 \to -$.

We denote this by the triple $(C, \otimes, 1)$ where we omit the unitors.

Definition 4.2 (Semisimple Category)**.** An abelian category C is semisimple if every object is either simple or a direct sum of simple objects. An object is *simple* if its only subobjects are 0 or X itself. Any object can be written as

$$X = \bigoplus_{i \in I} S_i$$

Definition 4.3 (Abelian Category)**.**

Lemma 4.4 (Schurs 1st lemma (which one?))**.** If S and T are simple objects in a semisimple category over an algebraically closed field (this is always C in our case), then it holds that

$$\mathrm{Hom}(S, T) = \begin{cases} k, & \text{if } S = T; \\ 0, & \text{if } S \neq T. \end{cases}$$

Lets give a coordinatized approach to fusion categories:

Definition 4.5 (Fusion Category)**.** A *fusion category* is a finite set of labels (sectors?) L and vector spaces V_c^{ab} for all $a, b, c \in L$, as well as the isomorphisms

$$F_d^{abc} : \bigoplus_e V_e^{ab} \otimes V_d^{ec} \to \bigoplus_e V_d^{ae} \otimes V_e^{bc}$$

such that the pentagon holds.

Lemma 4.6. Let C be a k-linear monoidal category whose tensor product is k-linear in each variable. Then

$$\bigotimes_{r=1}^{n} \left(\bigoplus_{a_r} X_{r,a_r} \right) \cong \bigoplus_{\hat{a}} \bigotimes_{r=1}^{n} X_{r,a_r},$$

where

$$\hat{a} = (a_1, \ldots, a_n).$$

Lemma 4.7 (Finite Groups admit Fusion Category)**.** Let G be a finite group, the $\mathbf{Rep}(G)$ is a fusion category.

Let $\mathrm{Vec} := (\mathrm{FD}\mathbb{C}\mathrm{Vec}, \otimes, \mathbb{C})$ be the monoidal category of finite dimensional complex vector spaces where the last tuple entry $\mathbb{C}$ denotes the unit object, since there are isomorphisms $V \otimes \mathbb{C} \cong V \cong C \otimes V$.

5 implementation

We want to be able to represent a traditional tensor op such as

```
np.einsum("ijklmn,nkol -> ijmo", a, b)
```

for symmetric tensors. As we want to reuse code form existing librariries that is written using numpy, it would be ideal to write our classes in a way that would only require substituting *np* for our own libraries *snp*. Towards this goal we define a abstractclass Tensor

```
class Tensor(abc.ABC):
  ...
  @abstractmethod
  def transpose(cls, new_shape:list[int]): ...
  ...
```

and we implement both an *np* wrapper $NPTensor$ and our symmetric Tensor as $STensor$, i.e.

```
class STensor(Tensor):

  def __init__(self, ...):
    self.open_legs:list[int|str]
    self.fusion_tree:FusionTree(self.open_legs, ...)
    ...
  @abstractmethod
  def transpose(cls, new_shape:list[int]): ...
  ...
```

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